What is PostgreSQL?

Instance Running server.

- **Cluster** Collection of databases.
  - **Roles** Control access.
  - **Database** Logical isolation.
    - **Schema** Object namespace.

Tables	Indexes	Sequences	Routines

What we use at SumUp

Classic

RDS PostgreSQL 15.18

Fleet

Aurora PostgreSQL 18.6

Storage model

Pages	Storage units. 8 kB blocks; a table or an index is a file of pages.
Tuples	Row versions. UPDATE = DELETE + INSERT
TOAST	Large-value storage. Oversized values are compressed and moved out of line into a TOAST table.

Storage model

Table = heap file

a sequence of 8 kB pages

- page 0
- page 1
- page 2
- page 3

...

Reads and writes move whole pages between disk and shared buffers.

One page, zoomed in

tuples are row versions

- page header
- line pointers → where each tuple starts
- free space
- tuple · row 42, version 1 · dead until VACUUM
- tuple · row 42, version 2 · live
- tuple · row 7 · live

UPDATE never overwrites: it adds a new version and leaves the old one for VACUUM.

TOAST (The Oversized-Attribute Storage Technique)

oversized values leave the page

tuple · row 9
description = pointer to TOAST

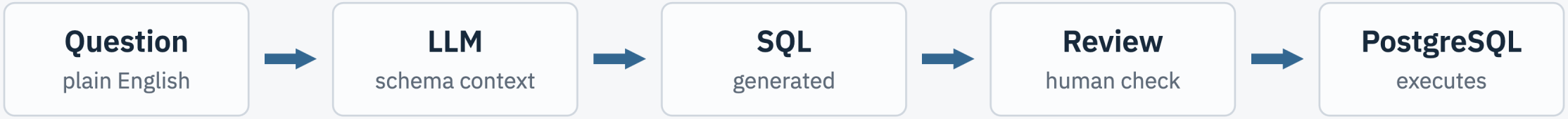
`pg_toast.pg_toast_<oid>`

- chunk 0 · compressed
- chunk 1
- chunk 2 ...

Values past ~2 kB are compressed and split into ~2 kB chunks in a separate table.

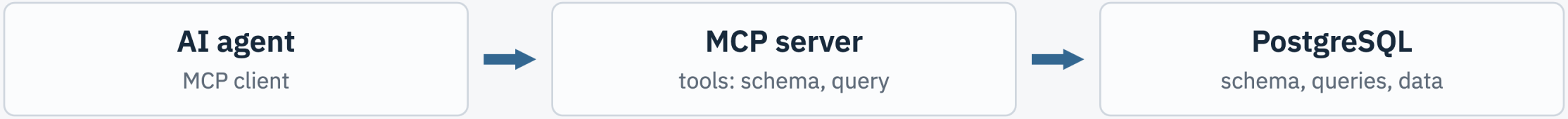
PostgreSQL and AI

NL2SQL Generate SQL from questions.



Google Cloud: NL2SQL with BigQuery and Gemini

MCP Expose database tools to AI agents.



modelcontextprotocol.io

Other major cloud offerings






```
SELECT name, review
FROM reviews
WHERE ai.if(
  'Is this review positive? ' || review
);
```

more AI functions






```
SELECT azure_ai.generate(
  'Summarize in 10 words: ' || review
) AS summary
FROM reviews
WHERE azure_ai.is_true(
  'This review is positive: ' || review
);
```

more AI functions

pg_*

Catalogs, system views, extensions

System catalogs and views

`pg_class`	Relations.
`pg_attribute`	Column metadata.
`pg_type`	Data types.
`pg_indexes`	Index definitions.
`pg_database`	Database metadata.
`pg_roles`	Permissions and roles.
`pg_extension`	Installed.
`pg_available_extensions`	Installable.
`pg_stats`	Planner stats.
`pg_stat_activity`	Live sessions.

Extensions: add new capabilities

`pg_surgery`
Low-level data repair. Emergency use only.

`pg_trgm`
Fuzzy matching. Similarity search.

```
-- install (needs the files on the server)
CREATE EXTENSION IF NOT EXISTS pg_trgm;
```

`pg_cron`
Schedule SQL jobs inside PostgreSQL.

`pgvector`
Vector storage. Semantic search.

OFFICIALLY SUPPORTED LANGUAGES

- **C** — the native API, built with PGXS.
- **SQL and PL/pgSQL** — pure SQL extensions.
- **PL/Python, PL/Perl, PL/Tcl** — procedural languages shipped with PostgreSQL.

Demo time

A practical example: how to build a PostgreSQL extension in Go.

PostgreSQL tools

pganalyze	Understand what is slow and why. Query insights, explain plans, index and vacuum advice.
DBtune	Move from monitoring to optimizing. AI-driven tuning of parameters, indexes and autovacuum.
pgBadger	Turn PostgreSQL logs into reports. Slow queries, errors, locks and checkpoints, read from the log files.

pganalyze

Understand what is slow and why.

pganalyze.com

DBtune

Move from monitoring problems to optimizing the database.

dbtune.com

pgBadger

Turn PostgreSQL logs into reports.

https://dbops-pgbadger.fleet.live.sumup.net/fleet/

Installed in all our clusters.

My favorite talk

What Is NULL?

Vik Fearing

Read the original:

thegresqlpost.org/presentations/what-is-null/2026-10-21

PostgreSQL is not just a database. It’s an extensible platform.

One engine, many specialized systems:

PostGIS
Geospatial database.

pgvector
Vector similarity search.

TimescaleDB
Time-series database.

Apache AGE
Graph queries, openCypher.

ParadeDB
Full-text search, BM25.